

homework5

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We continue working with the World Top Incomes Database [<https://wid.world>], and the Pareto distribution, as in the lab. We also continue to practice working with data frames, manipulating data from one format to another, and writing functions to automate repetitive tasks.

We saw in the lab that if the upper tail of the income distribution followed a perfect Pareto distribution, then

$$\left(\frac{P99}{P99.9}\right)^{-a+1} = 10 \quad (1)$$

$$\left(\frac{P99.5}{P99.9}\right)^{-a+1} = 5 \quad (2)$$

$$\left(\frac{P99}{P99.5}\right)^{-a+1} = 2 \quad (3)$$

We could estimate the Pareto exponent by solving any one of these equations for a ; in lab we used

$$a = 1 - \frac{\log 10}{\log (P99/P99.9)} , \quad (4)$$

Because of measurement error and sampling noise, we can't find one value of a which will work for all three equations (1)–(3). Generally, trying to make all three equations come close to balancing gives a better estimate of a than just solving one of them. (This is analogous to finding the slope and intercept of a regression line by trying to come close to all the points in a scatterplot, and not just running a line through two of them.)

1.

We estimate a by minimizing

$$\left(\left(\frac{P99}{P99.9}\right)^{-a+1} - 10\right)^2 + \left(\left(\frac{P99.5}{P99.9}\right)^{-a+1} - 5\right)^2 + \left(\left(\frac{P99}{P99.5}\right)^{-a+1} - 2\right)^2$$

Write a function, `percentile_ratio_discrepancies`, which takes as inputs `P99`, `P99.5`, `P99.9` and `a`, and returns the value of the expression above. Check that when `P99=1e6`, `P99.5=2e6`, `P99.9=1e7` and `a=2`, your function returns 0.

用以下代码完成

```

# 问题 1: 定义计算分位数比率差异平方和的函数
percentile_ratio_discrepancies <- function(a, P99, P99.5, P99.9) {
  # 计算三个分位数比率
  ratio1 <- P99 / P99.9
  ratio2 <- P99.5 / P99.9
  ratio3 <- P99 / P99.5

  # 计算理论上的指数关系
  term1 <- (ratio1^(-a + 1) - 10)^2
  term2 <- (ratio2^(-a + 1) - 5)^2
  term3 <- (ratio3^(-a + 1) - 2)^2

  # 返回总差异平方和
  return(term1 + term2 + term3)
}

# 验证函数: 当输入特定值时, 结果应为 0
test_val <- percentile_ratio_discrepancies(a = 2, P99 = 1e6, P99.5 = 2e6, P99.9 = 1e7)
cat(" 验证函数结果 (应为 0) :", test_val, "\n")

```

验证函数结果 (应为0) : 0

2.

Write a function, `exponent.multi_ratios_est`, which takes as inputs `P99`, `P99.5`, `P99.9`, and estimates `a`. It should minimize your `percentile_ratio_discrepancies` function. The starting value for the minimization should come from (4). Check that when `P99=1e6`, `P99.5=2e6` and `P99.9=1e7`, your function returns an `a` of 2.

用以下代码解决:

```

# 问题 2: 定义通过最小化差异估计 a 的函数
exponent.multi_ratios_est <- function(P99, P99.5, P99.9) {
  # 使用公式 (4) 计算初始值
  initial_a <- 1 - log10(10) / log10(P99 / P99.9)

  # 优化函数: 在合理范围内寻找最小化差异的 a
  result <- optimize(
    f = function(a) percentile_ratio_discrepancies(a, P99, P99.5, P99.9),
    interval = c(0.5, 5), # a 的合理搜索范围
    tol = 1e-8
  )
}

```

```

)

return(result$minimum)
}

# 验证函数：当输入特定值时，结果应为 2
test_est <- exponent.multi_ratios_est(P99 = 1e6, P99.5 = 2e6, P99.9 = 1e7)
cat(" 验证估计结果（应为 2）:", test_est, "\n")

```

```
## 验证估计结果（应为2）：2
```

3.

Write a function which uses `exponent.multi_ratios_est` to estimate a for the US for every year from 1913 to 2012. (There are many ways you could do thi, including loops.) Plot the estimates; make sure the labels of the plot are appropriate.

用以下代码解决：

```

# 读取数据（注意调整路径以匹配实际位置）
wtid_data <- read_csv("data/wtid-report.csv") %>%
  # 重命名列以方便使用
  rename(
    P90 = `P90 income threshold`,
    P95 = `P95 income threshold`,
    P99 = `P99 income threshold`,
    P99.5 = `P99.5 income threshold`,
    P99.9 = `P99.9 income threshold`,
    P99.99 = `P99.99 income threshold`
  ) %>%
  # 筛选美国数据（尽管数据中只有美国）
  filter(Country == "United States") %>%
  # 选择所需年份（1913-2012）
  filter(Year >= 1913 & Year <= 2012)

```

```

## Rows: 100 Columns: 8
## -- Column specification -----
## Delimiter: ","
## chr (1): Country
## dbl (7): Year, P90 income threshold, P95 income threshold, P99 income thresh...

```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# 问题 3: 为每年估计 Pareto 指数 (多比率方法)
```

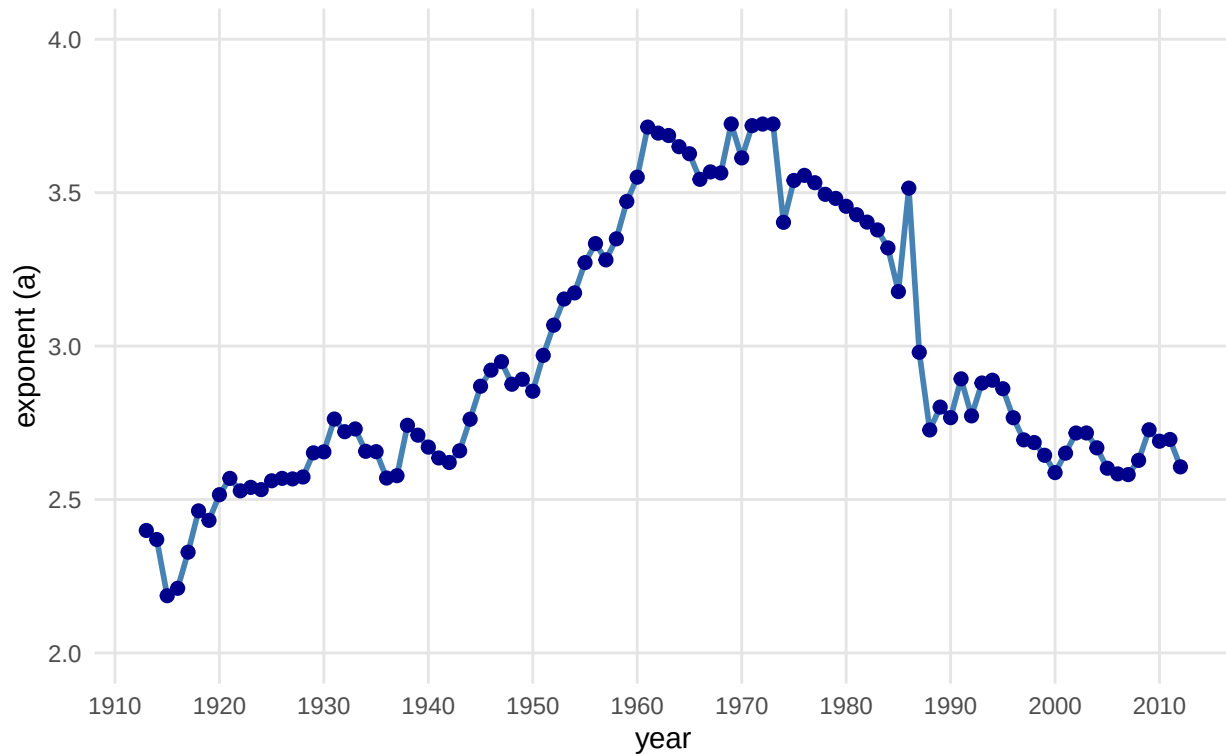
```
wtid_data <- wtid_data %>%
  rowwise() %>%
  mutate(
    a_multi = exponent_multi_ratios_est(P99, P99.5, P99.9)
  ) %>%
  ungroup()
```

```
# 绘制 Pareto 指数随时间变化图
```

```
ggplot(wtid_data, aes(x = Year, y = a_multi)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(size = 2, color = "darkblue") +
  labs(
    title = "Estimation of the Pareto Index in the United States(1913-2012)",
    subtitle = "Based on the multi-quantile ratio method",
    x = "year",
    y = "exponent (a)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    plot.subtitle = element_text(hjust = 0.5),
    panel.grid.major = element_line(color = "grey90"),
    panel.grid.minor = element_blank()
  ) +
  scale_x_continuous(breaks = seq(1910, 2010, 10)) +
  scale_y_continuous(limits = c(2.0, 4.0), breaks = seq(2.0, 4.0, 0.5))
```

Estimation of the Pareto Index in the United States(1913–2012)

Based on the multi-quantile ratio method



4.

Use (4) to estimate a for the US for every year. Make a scatter-plot of these estimates against those from problem 3. If they are identical or completely independent, something is wrong with at least one part of your code. Otherwise, can you say anything about how the two estimates compare?

用以下代码解决：

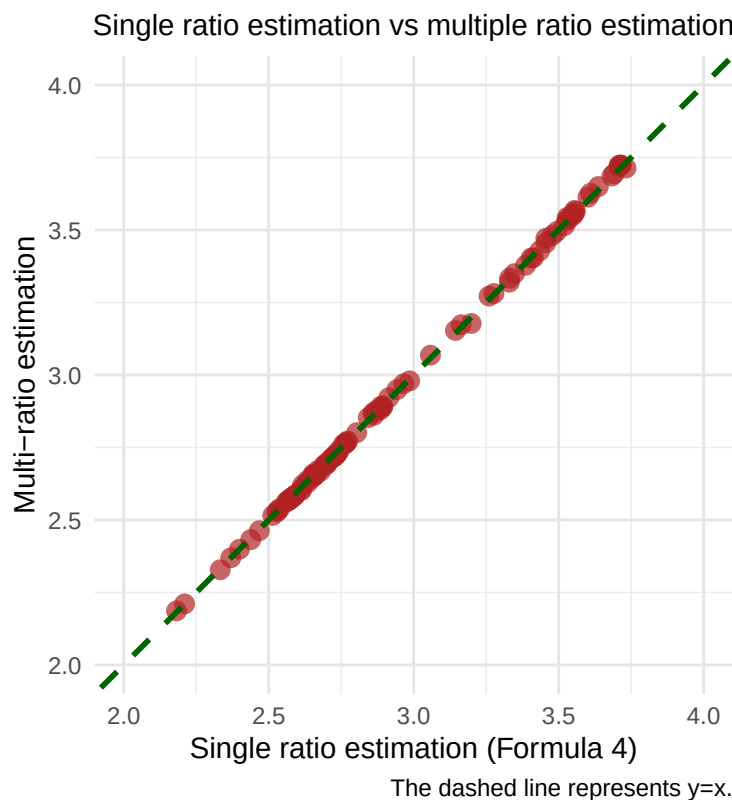
```
wtid_data <- wtid_data %>%  
  mutate(  
    a_single = 1 - log10(10) / log10(P99 / P99.9)  
  )  
  
# 绘制两种估计方法的比较散点图  
ggplot(wtid_data, aes(x = a_single, y = a_multi)) +  
  geom_point(size = 3, alpha = 0.7, color = "firebrick") +  
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "darkgreen", size = 1) +  
  labs(  
    title = "Comparison of Two Methods for Estimating the Pareto Index",  
    subtitle = "Single ratio estimation vs multiple ratio estimation",
```

```

x = "Single ratio estimation (Formula 4)",
y = "Multi-ratio estimation",
caption = "The dashed line represents y=x."
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  plot.subtitle = element_text(hjust = 0.5),
  panel.grid.major = element_line(color = "grey90")
) +
scale_x_continuous(limits = c(2.0, 4.0), breaks = seq(2.0, 4.0, 0.5)) +
scale_y_continuous(limits = c(2.0, 4.0), breaks = seq(2.0, 4.0, 0.5)) +
coord_fixed(ratio = 1)

```

Comparison of Two Methods for Estimating the Pareto Index



可选：展示数据表格

```
head(wtid_data)
```

```
## # A tibble: 6 x 10
```

```
##   Country      Year   P90   P95   P99  P99.5  P99.9 P99.99 a_multi a_single
```

##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	United States	1913	NA	NA	80088.	1.31e5	4.15e5	1.75e6	2.40	2.40
## 2	United States	1914	NA	NA	74013.	1.23e5	3.98e5	1.74e6	2.37	2.37
## 3	United States	1915	NA	NA	62392.	1.19e5	4.38e5	2.22e6	2.19	2.18
## 4	United States	1916	NA	NA	74869.	1.34e5	5.02e5	2.75e6	2.21	2.21
## 5	United States	1917	27009.	37405.	92341.	1.50e5	5.20e5	2.37e6	2.33	2.33
## 6	United States	1918	28584.	38189.	92221.	1.43e5	4.43e5	1.79e6	2.46	2.47